

Indian Statistical Institute, Bangalore

B. Math.

First Year, First Semester

Analysis I

Final Examination

Total Marks 7x15=105

Maximum marks: 100

Instructor: B V Rajarama Bhat

Date : Nov. 17, 2025

Time: 3 hours

- (1) Let $s : [0, 1] \rightarrow [0, \infty)$ be a function. Suppose

$$\sum_{a \in A} s(a) \leq 100,$$

for every finite subset A of $[0, 1]$. Show that the set

$$B = \{x \in [0, 1] : s(x) \neq 0\}$$

is countable. (Hint: First show that for every $n \in \mathbb{N}$ the set $\{x \in [0, 1] : s(x) > \frac{1}{n}\}$ is finite.) [15]

- (2) Prove or disprove: (i) If $\{a_n\}_{n \in \mathbb{N}}$ and $\{b_n\}_{n \in \mathbb{N}}$ are convergent sequences of real numbers then

$$\lim_{n \rightarrow \infty} (a_n b_n) = \left(\lim_{n \rightarrow \infty} a_n \right) \left(\lim_{n \rightarrow \infty} b_n \right).$$

(ii) If $\{a_n\}_{n \in \mathbb{N}}$ and $\{b_n\}_{n \in \mathbb{N}}$ are bounded sequences of real numbers, then

$$\limsup_{n \rightarrow \infty} (a_n b_n) \leq \left(\limsup_{n \rightarrow \infty} a_n \right) \left(\limsup_{n \rightarrow \infty} b_n \right).$$

- (3) Let $\{x_n\}_{n \in \mathbb{N}}$ is a sequence of real numbers such that $\sum_{n=1}^{\infty} \frac{x_n}{n}$ converges. Show that [15]

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n x_k = 0.$$

- (4) (i) Let $g : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Show that g is uniformly continuous. (ii) Give an example of a continuous function $h : [0, 1) \rightarrow \mathbb{R}$ which is not uniformly continuous. [15]

- (5) Let a, b be real numbers and let $m : \mathbb{R} \rightarrow \mathbb{R}$ be the function

$$m(x) = ax + b, \quad x \in \mathbb{R}.$$

Suppose $h : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function satisfying

$$h(x) = h(m(x)), \quad \forall x \in \mathbb{R}.$$

Under what conditions on a, b , the function h is forced to be a constant function? Prove your claim. [15]

- (6) Let $a_1 < b_1$, and $a_2 < b_2$ be real numbers. Suppose $f : [a_1, b_1] \rightarrow [a_2, b_2]$ and $g : [a_2, b_2] \rightarrow \mathbb{R}$ be functions. Suppose $c \in [a_1, b_1]$ and $d = f(c)$. Assume that f is differentiable at c and g is differentiable at d . Show that $g \circ f$ is differentiable at c and

$$(g \circ f)'(c) = g'(d)f'(c).$$

- (7) Suppose $f : [0, 1] \rightarrow \mathbb{R}$ is continuous on $[0, 1]$ and is differentiable on $(0, 1)$. Suppose $\lim_{x \rightarrow 1^-} f'(x)$ exists and equals z . Show that f is differentiable at 1 and $f'(1) = z$. [15]